

Personal Project

Loan Approval Analysis: Identifying Key Drivers of Loan Approval Decisions

Tools: SQL, Power BI, Excel

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Date: 06/ 2026

1. Business Problem

Loan approval is a critical process for financial institutions because it directly affects both profitability and risk exposure. Approving high-risk applicants may increase default rates, while rejecting qualified borrowers may result in missed business opportunities.

The objective of this project is to analyze loan application data and identify the key factors influencing loan approval decisions. Through SQL analysis and Power BI visualization, this study aims to uncover approval patterns, segment customers based on financial characteristics, and provide actionable recommendations to improve lending strategies.

Key business questions include:

- 1. What factors have the strongest impact on loan approval decisions?
- 2. Which customer segments have the highest approval potential?
- 3. What actions can lenders take to improve approval efficiency while managing risk?

2. Dataset Overview

The dataset used in this project is the **Financial Risk for Loan Approval** dataset obtained from Kaggle. It contains **20,000 loan application records** and **36 variables** describing applicants' demographic characteristics, employment status, financial condition, credit profile, and loan-related information.

The primary objective of the dataset is to evaluate the factors influencing loan approval decisions. Therefore, the analysis focuses on the target variable **LoanApproved**, which indicates whether a loan application was approved (1) or rejected (0).

To answer the business questions, several key variables were selected for analysis, including income level, debt burden, credit score, home ownership status, employment status, and loan purpose. These variables were further transformed into custom bands and customer segments to facilitate interpretation and visualization.

Key Variables Used in the Analysis:

Feature	Type	Description
AnnualIncome	Numeric	Applicant's annual income
TotalDebtToIncomeRatio	Numeric	Total debt burden relative to income
CreditScore	Numeric	Applicant's creditworthiness score

HomeOwnershipStatus	Categorical	Housing ownership status (Own, Rent, Mortgage, Other)
EmploymentStatus	Categorical	Employment situation of the applicant
LoanPurpose	Categorical	Purpose of the loan application
LoanApproved	Binary	Target variable indicating loan approval outcome
EducationLevel	Categorical	Highest level of education attained

3. Methodology

This project follows a descriptive and exploratory data analysis approach to identify the key factors influencing loan approval decisions and to uncover meaningful borrower segments.

The analysis was conducted in three four stages:

3.1 Data Preparation

The dataset was first cleaned and validated to ensure consistency and accuracy. Data types were reviewed and converted where necessary to support calculations and aggregation. Several derived variables were created to facilitate business analysis, including:

- Income Bands based on annual income levels
- Debt-to-Income (DTI) Bands based on borrowers' debt burden
- Credit Score Bands based on creditworthiness levels

3.2 Exploratory Data Analysis (EDA)

Exploratory analysis was performed using SQL to understand the overall characteristics of the dataset and identify patterns in loan approval outcomes.

3.3 Driver Analysis and Customer Segmentation

To determine the factors associated with loan approval, approval rates were compared across multiple dimensions. In addition, borrowers were grouped into customer personas to evaluate approval performance among different risk profiles. This segmentation approach helps translate analytical findings into actionable business insights.

3.4 Data Visualization

Power BI was used to create an interactive dashboard consisting of four sections:

- Executive Overview
- Approval Drivers Analysis
- Customer Personas
- Business Recommendations

The dashboard enables stakeholders to monitor key metrics, explore approval patterns, and support data-driven lending decisions.

4. Key Findings

4.1 Debt-to-Income Ratio is the Strongest Approval Driver

The analysis reveals that Debt-to-Income Ratio (DTI) has the most significant impact on loan approval outcomes. Approval rates decline dramatically as debt burdens increase, falling from **63.15%** among applicants with DTI **below 20%** to only **0.09%** among people with DTI **above 100%**

This pattern indicates that borrowers with lower debt burdens are substantially more likely to receive loan approval.

4.2 Income Positively Influences Approval Probability

Higher income levels are associated with significantly higher approval rates for loans (46.29%). Applicants with income above 70K have a 72 times higher approval rate compared to those with low income (0-30K). Higher-income applicants demonstrate stronger repayment capacity and therefore achieve significantly higher approval rates.

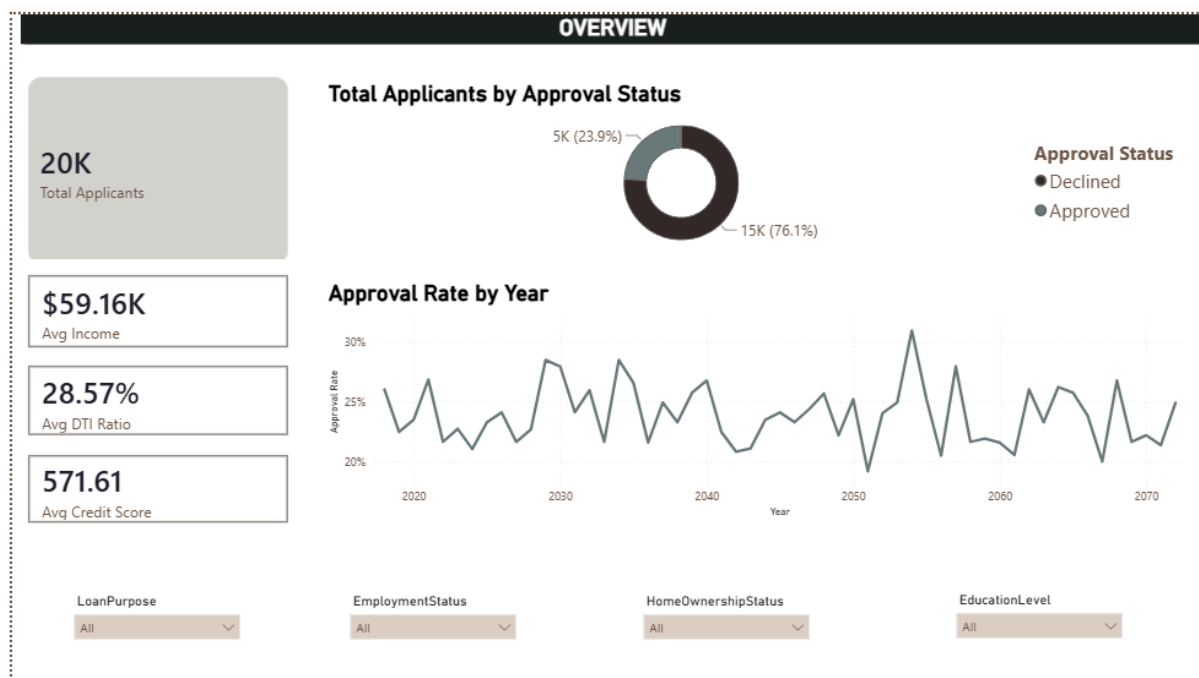
4.3 Customer Personas Reveal Distinct Risk Profiles

Three customer personas were identified:

- Prime Borrowers: 73.71% approval rate
- Emerging Borrowers: 25.15% approval rate
- High-Risk Borrowers: 0.38% approval rate

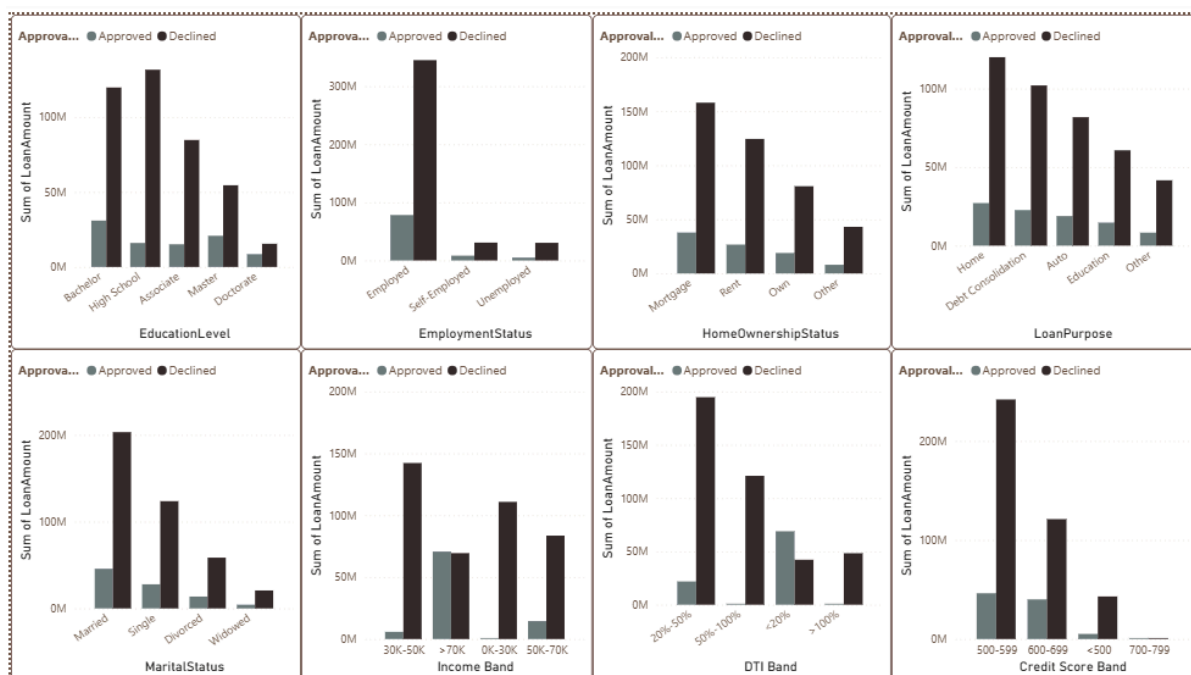
Prime Borrowers represent the most valuable customer segment, while High-Risk Borrowers exhibit extremely low approval likelihood.

5. Visualization



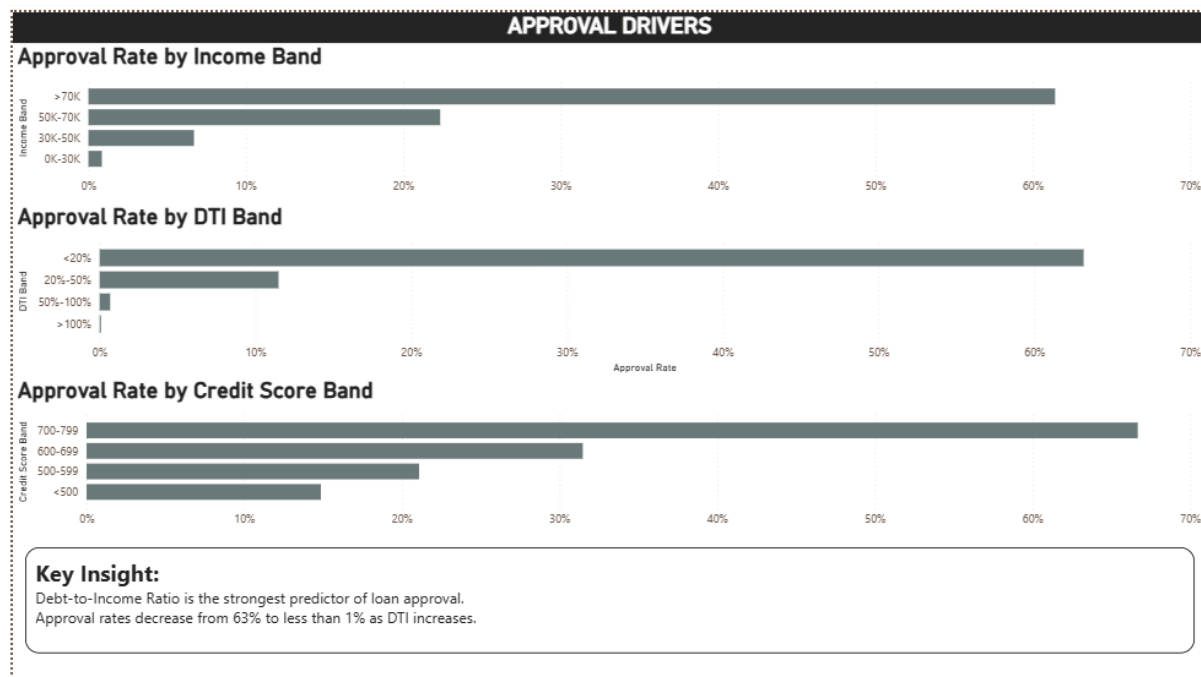
Out of **20,000 loan applications**, only **4,780 applications (23.9%)** were approved, while **76.1% were rejected**. This indicates that the lender follows a relatively strict approval process to manage credit risk and maintain portfolio quality.

These figures suggest that most applicants belong to the middle-income segment with average creditworthiness rather than premium borrower groups.



Approved loans are generally associated with borrowers who have: Higher income levels; better credit scores; lower debt burdens. This suggests that financial capacity plays a crucial role in loan approval decisions.

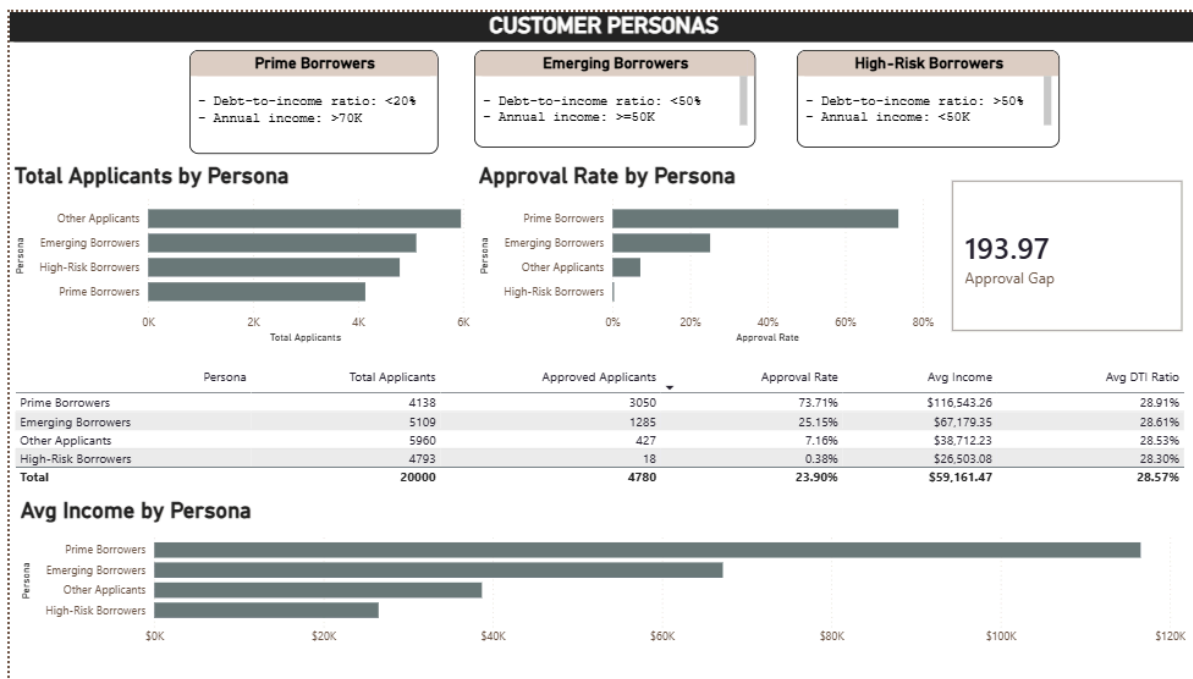
Applicants with **Mortgage** home ownership status account for the largest share of loan applications and approved loans, making them a key customer group within the lending portfolio.



Debt-to-income ratio appears to be a strong predictor of loan approval outcomes. Approval rates decline dramatically as debt burdens increase, falling from 63.15% among applicants with DTI below 20% to only 0.09% among people with DTI above 100%

Higher income levels are associated with significantly higher approval rates for loans (46.29%). Applicants with income above 70K have a 72 times higher approval rate compared to those with low income (0-30K)

The majority of applicants have credit score below 700. Loan approval rates increase significantly with credit score, with applicants with score above 700 demonstrate much higher approval rates (66.67%), but only a small portion of applicants (0.03%) have this range of credit score



Prime Borrowers achieve a 73.71% approval rate and represent the most valuable customer segment. Prime Borrowers are 194 times more likely to be approved than High-Risk Borrowers

Emerging Borrowers show moderate approval potential and may benefit from tailored loan products.

High-Risk Borrowers have an approval rate below 1%, driven by low income and high debt burden.

Other Applicants account for the largest share of applications, suggesting opportunities for further segmentation.

6. Business Recommendations

Recommendation 1: Prioritize Low-DTI Applicants

Implement automated screening processes that prioritize applicants with low debt-to-income ratios and flag high-DTI applications for additional review.

Recommendation 2: Fast-Track High-Income Borrowers

Develop streamlined approval procedures for high-income applicants to improve customer experience and operational efficiency.

Recommendation 3: Expand Opportunities for Emerging Borrowers

Introduce customized loan products and flexible lending options to increase approval opportunities for borrowers with moderate risk profiles.

Recommendation 4: Strengthen Risk Controls

Apply stricter underwriting standards and enhanced verification processes for high-risk applicants to reduce potential credit losses.